Design methods for composite X and Hadamard gates

Hristo G. Tonchev and Nikolay V. Vitanov

The design conditions are summarized in Table 1. Here $a_k = \Omega_k/\Omega_0$, $p = \max_k a_k$, and $A = \Phi/\pi = \sum_k a_k$ for equal-duration segments. We write B_r^{n} for the square in (ϵ, δ) and B_r for the square in (ϵ, δ_c), where $\delta_c = \Delta/\Omega_{\max} = \delta/p$. A superscript n or c on a mean or threshold fraction specifies these two coordinates. The threshold fraction f_4 counts samples with $\mathcal{I} \le 10^{-4}$; $f_4^{(0)}$ restricts it to the connected component containing the nominal point. In the shorter coverage searches, a smooth approximation to f_4 is maximized, rather than minimizing J_w:

$$S_s(Q) = \frac{1}{M} \sum_{q \in Q} \left[1 + e^{[\mathcal{I}(q) - 10^{-4}]/s}\right]^{-1}, \qquad s > 0.$$

All averages and protection conditions are sampled; they do not certify continuum bounds.

Table 1: Design classes and selection rules. Numerical resource limits are in the $\Omega_0 T = \pi$ convention; approximate values are rounded, with exact bounds and sample sets in the repository. Exact nominal matching is imposed on the full rectangular-pulse unitary up to global phase, not on the finite-edge unitary. All edge-aware rows use common-peak coordinates, the 4 ns envelope and weights specified above. The paired bounds are $(A_c, p_c) \simeq (13, 1.13136)$, $(15, 1.14030)$, $(10.32419, .96755)$ and $(10.98658, .84092)$ for X13w, X15w, H13w and H15w, respectively.

Controls	Objective, model and error domain	Nominal target and resource bounds	Central protection	Candidate selection
Symmetric X_{na}, X_{nb}	Cancel the propagator derivatives specified in the X-gate section of the manuscript; local expansion in (ϵ, δ), rectangular pulses.	Target up to global phase; $a_k = 1$, $A = N$, $p = 1$.	Specified pure and mixed derivative conditions; longer numerical roots satisfy them approximately.	Retain the distinct phase branches described in the corresponding subsections.
X5c	Maximize smooth $f_4^{\mathrm{n}}(0.15)$; rectangular pulses.	$\mathcal{I}_0 \lesssim 8.272 \times 10^{-5}$, not exact matching; $10^{-5} \le a_k \le 2$, $A \le 6.9933$.	$\overline{\mathcal{I}}^{\mathrm{n}}_{.15} \lesssim 2.482 \times 10^{-4}$; $\overline{\mathcal{I}}^{\mathrm{c}}_{.15} \lesssim 2.099 \times 10^{-3}$.	Increased threshold fraction and reduced nominal error within the resource and mean-error budgets.
X7c	Maximize the equal-weight smooth fractions on $B^{\mathrm{n}}_{.5}$ and $B_{.5}$; rectangular pulses.	Exact nominal target; $10^{-5} \le a_k \le 1.1412$, $A \le 6.9583$.	$\overline{\mathcal{I}}^{\mathrm{n}}_{.15} \lesssim 5.044 \times 10^{-5}$, $\overline{\mathcal{I}}^{\mathrm{c}}_{.15} \lesssim 6.234 \times 10^{-5}$.	Rank by the smaller relative gain in the two wide-domain fractions; require nondecreasing small/wide fractions and nonincreasing means in both coordinates, including Sobol means.
X11c	Same two-coordinate coverage objective as X7c.	Exact nominal target; $10^{-5} \le a_k \le 1.1016$, $A \le 10.8489$.	Means $\lesssim (6.925, 8.055) \times 10^{-6}$ in n,c coordinates; sampled maxima $\le (4.122, 4.883) \times 10^{-5}$.	Same acceptance and ranking as X7c; the search additionally bounds the sampled central maxima.
X9c	Minimize $\frac{1}{2}(\overline{\mathcal{I}}^{\mathrm{n}}_{.15} + \overline{\mathcal{I}}^{\mathrm{c}}_{.15})$; rectangular pulses.	Exact nominal target; $.05 \le a_k \lesssim 1.9451$, $A \lesssim 11.6385$.	No additional central threshold condition.	Lowest two-coordinate mean among feasible converged candidates on a finer grid.
X13c, X15c	Same two-coordinate mean objective as X9c.	Exact nominal target; $.05 \le a_k \le 2$, $A \le N$.	No additional central threshold condition.	Lowest two-coordinate mean among feasible polished candidates; require finer-grid and separate Sobol agreement.
X13w, X15w	Minimize $J_w(B_{.5})$; rectangular pulses.	Exact nominal target; $.05 \le a_k \le p_c$, $A \le A_c$, using the paired c control.	Sampled smooth upper bound on $\mathcal{I}/10^{-4}$ is at most one on $B_{.20}$.	Largest connected $f_4^{(0)}(.5)$ among feasible candidates; verify the protected square and resource limits on finer samples.
H5c	Maximize smooth $f_4^{\mathrm{n}}(.5)$ after two-coordinate mean refinement; rectangular pulses only.	Exact nominal target; $10^{-5} \le a_k \lesssim 1.9924$, $A \lesssim 7.5466$.	Means $\lesssim 1.179 \times 10^{-4}$ (n), 9.855×10^{-4} (c); small quadrature allowances enter selection.	Largest wide-domain n-coordinate threshold fraction among qualifying candidates, with fine-grid and resource checks.
H7c, H11c	Minimize $.3J_{w,\mathrm{rect}}(B_{.5}) + .7J_{w,\mathrm{edge}}(B_{.5})$ after weighted-mean refinement.	Exact nominal target; $a_k \ge .05$. $(A_{\max}, p_{\max}) \simeq (7.65996, 1.32562)$, $(10.12581, 1.15353)$.	Weighted central mean $J_c \lesssim 3.828 \times 10^{-4}$, 3.787×10^{-5}, respectively.	Largest $.3f_{4,\mathrm{rect}}(.5) + .7f_{4,\mathrm{edge+noise}}(.5)$ among feasible candidates.

Controls	Objective, model and error domain	Nominal target and resource bounds	Central protection	Candidate selection
H9c; H13c, H15c	Minimize J_c on $B_{.15}$.	Exact nominal target; $a_k \geq .05$. $(A_{\max}, p_{\max}) \simeq (6.93220, .86919)$, $(13, 2)$, $(10.98658, .95862)$.	No additional central threshold condition.	Lowest $.3\overline{\mathcal{I}}_{\mathrm{rect},.15} + .7\overline{\mathcal{I}}_{\mathrm{edge+noise},.15}$ among feasible candidates.
H13w	Minimize $.3J_{w,\mathrm{rect}}(B_{.5}) + .7J_{w,\mathrm{edge}}(B_{.5})$.	Exact nominal target; $.05 \leq a_k \leq p_c$, $A \leq A_c$.	Sampled smooth upper bound on $\mathcal{I}/10^{-4}$ is at most one for both models on $B_{.15}$.	Largest weighted rectangular/noisy-edge threshold fraction, as for H7c/H11c.
H15w	Same weighted width objective as H13w, followed by finer central-constraint refinement.	Exact nominal target; $.05 \leq a_k \leq p_c$, $A \leq A_c$.	Same two-model condition on $B_{.20}$.	Retain the feasible refined control; its weighted rectangular/noisy-edge coverage is evaluated after refinement.

Noise enters the candidate selection, rather than the deterministic optimization, for the edge-aware Hadamard rows of Table 1. The selection uses finite-band Gaussian $1/f$ detuning noise with $S_\xi(f) = A/f$, $A = 3.6 \times 10^5\,\mathrm{Hz}^2$, on 1 Hz–1 MHz. The same fixed ensemble is used across candidates: 16 trajectories comprising eight independent antithetic pairs. The subsequent fixed-coefficient tests in the finite-edge appendix of the manuscript use fresh Gaussian coefficients, with separate ensembles for the domain comparison and the nominal bandwidth sweep. Within that sweep, the Gaussian coefficients are reused across cutoffs; the cutoff points are therefore correlated. H5c and the numerical X controls are selected without dynamic noise. Exact meshes, smoothing parameters, bounds, starts and search settings are supplied in the repository.